

Workshop: Differentiation Product and Quotient Rules

UEA LEARNING ENHANCEMENT TEAM MATHEMATICS & STATISTICS

Product Rule

Functions that are made of other functions multiplied together can be differentiated using the product rule.

$$y = f(x) \times g(x)$$

For example, you may have the function $y = x^2 e^x$.

Use 'dummy' variables to represent the two functions:

$$u = f(x), \quad v = g(x)$$

Then:

$$\frac{dy}{dx} = v \frac{du}{dx} + u \frac{dv}{dx}$$

So, in the example:

$$\begin{aligned} u &= x^2, & v &= e^x \\ \frac{du}{dx} &= 2x, & \frac{dv}{dx} &= e^x \\ \frac{dy}{dx} &= 2xe^x + x^2 e^x = xe^x(2 + x) \end{aligned}$$

Differentiate the following:

1. $y = 5xe^x$

$$\begin{aligned} u &= 5x, & v &= e^x \\ u' &= 5, & v' &= e^x \\ \frac{dy}{dx} &= 5e^x + 5xe^x = 5e^x(1 + x) \end{aligned}$$

2. $y = x^3 \ln x$

$$u = x^3, \quad v = \ln x$$

$$u' = 3x^2, \quad v = \frac{1}{x}$$

$$\frac{dy}{dx} = x^2 + 3x^2 \ln x$$

3. $y = 4e^x \ln 4x$

$$u = 4e^x, \quad v = \ln 4x$$

$$u' = 4e^x, \quad v' = \frac{1}{x}$$

$$\frac{dy}{dx} = 4e^x \ln 4x + \frac{4e^x}{x} = 4e^x \left(\ln x + \frac{1}{x} \right)$$

4. $y = (6x^2 + 4x + 1) \ln x^2$

$$u = 6x^2 + 4x + 1, \quad v = \ln x^2$$

$$u' = 12x + 4, \quad v' = \frac{2}{x}$$

$$\frac{dy}{dx} = (12x + 4) \ln x^2 + \frac{2}{x} (6x^2 + 4x + 1)$$

5. $y = \left(5x - \frac{3}{x^2} \right) e^x$

$$u = 5x - 3x^{-2}, \quad v = e^x$$

$$u' = 5 + 6x^{-3}, \quad v' = e^x$$

$$\frac{dy}{dx} = \left(5 + \frac{6}{x^3} \right) e^x + \left(5x - \frac{3}{x^2} \right) e^x = e^x \left(5 + \frac{6}{x^3} + 5x - \frac{3}{x^2} \right)$$

Quotient Rule

When a function is produced by a function divided by another function, it can be differentiated with the quotient rule.

$$y = \frac{f(x)}{g(x)}$$

Let $u = f(x)$ and $v = g(x)$

$$\frac{dy}{dx} = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$$

For example: $y = \frac{x^2+5}{2x-1}$

$$u = x^2 + 5, \quad v = 2x - 1$$

$$\frac{du}{dx} = 2x, \quad \frac{dv}{dx} = 2$$

$$\frac{dy}{dx} = \frac{(2x-1) \times 2x - (x^2+5) \times 2}{(2x-1)^2} = \frac{4x^2 - 2x - 2x^2 + 10}{(2x-1)^2}$$

$$\frac{dy}{dx} = \frac{2x^2 - 2x + 10}{(2x-1)^2}$$

Differentiate the following:

6. $y = \frac{5x+2}{x-4}$

$$u = 5x + 2, \quad v = x - 4$$

$$u' = 5, \quad v' = 1$$

$$\frac{dy}{dx} = \frac{5(x-4) - 1(5x+2)}{(x-4)^2} = \frac{-22}{(x-4)^2}$$

7. $y = \frac{e^x}{x^2+2x}$

$$u = e^x, \quad v = x^2 + 2x$$

$$u' = e^x, \quad v' = 2x + 2$$

$$\frac{dy}{dx} = \frac{e^x(x^2+2x) - e^x(2x+2)}{(x^2+2x)^2} = \frac{e^x(x^2-2)}{(x^2+2x)^2}$$

8. $y = \frac{\ln 4x}{x^2+5}$

$$u = \ln 4x, \quad v = x^2 + 5$$

$$u' = \frac{1}{x}, \quad v' = 2x$$

$$\frac{dy}{dx} = \frac{\frac{1}{x}(x^2+5) - 2x \ln 4x}{(x^2+5)^2} = \frac{x^2+5-2x^2 \ln 4x}{x(x^2+5)^2}$$

9. $y = \frac{\sqrt{x}}{e^x}$

$$u = x^{\frac{1}{2}}, \quad v = e^x$$

$$u' = \frac{1}{2}x^{-\frac{1}{2}}, \quad v' = e^x$$

$$\frac{dy}{dx} = \frac{\frac{1}{2\sqrt{x}}e^x - e^x\sqrt{x}}{e^{2x}} = \frac{1-x}{2\sqrt{x}e^x}$$

Mixed

The following questions use product and quotient rule. They also make use of the chain rule. Find the derivative

$$10. y = e^{4x}(5x + 2)$$

$$u = e^{4x}, \quad v = 5x + 2$$

$$u' = 4e^{4x}, \quad v' = 5$$

$$y' = 4e^{4x}(5x + 2) + 5e^{4x} = e^{4x}(20x + 13)$$

$$11. y = (5x - 1)^4 \ln x$$

$$u = (5x - 1)^4, \quad v = \ln x$$

$$u' = 20(5x - 1)^3, \quad v' = \frac{1}{x}$$

$$y' = 20(5x - 1)^3 \ln x + \frac{(5x - 1)^4}{x} = \frac{(5x - 1)^3(20x \ln x + 5x - 1)}{x}$$

$$12. y = \frac{2x+5}{e^{9x}}$$

$$u = 2x + 5, \quad v = e^{9x}$$

$$u' = 2, \quad v' = 9e^{9x}$$

$$y' = \frac{2e^{9x} - 9e^{9x}(2x + 5)}{e^{18x}} = \frac{-43 - 18x}{e^{9x}}$$

$$13. y = (x^2 + 3x + 1)e^x \ln x$$

$$u = (x^2 + 3x + 1), \quad v = e^x \ln x$$

$$u' = 2x + 3, \quad v' = e^x \ln x + \frac{e^x}{x}$$

$$y' = (2x + 3)e^x \ln x + (x^2 + 3x + 1) \left(e^x \ln x + \frac{e^x}{x} \right)$$

$$14. y = \frac{5xe^x}{x+2}$$

$$u = 5xe^x, \quad v = x + 2$$

$$u' = 5e^x + 5xe^x, \quad v' = 1$$

$$y' = \frac{5e^x(1+x)(x+2) - 5xe^x}{(x+2)^2}$$

$$15. y = \frac{6x-7}{x^2+1} - 5x \ln(2x-5)$$

$$\begin{aligned} u &= 6x - 7, & v &= x^2 + 1 \\ u' &= 6, & v' &= 2x \end{aligned}$$

$$\begin{aligned} f &= 5x, & g &= \ln(2x-5) \\ f' &= 5, & g' &= \frac{2}{2x-5} \end{aligned}$$

$$y' = \frac{6(x^2+1) - 2x(6x-7)}{(x^2+1)^2} - 5 \ln(2x-5) - \frac{10x}{2x-5}$$